

Session 1 Solutions

Lecture 11

1. Calculate $5^7 \bmod 11$ by hand using repeated squaring and the homomorphism theorem. (5 to the power 7 mod 11). Verify that the calculations would be much harder if you left the mod 11 calculation to the end.

$$5^7 \bmod 11$$

$$5^2 = 25 \equiv 3$$

$$5^4 = 3^2 = 9$$

$$5^6 = 5^2 * 5^4 = 3 * 9 = 27 \equiv 5$$

$$5^7 = 5^6 * 5 \equiv 5 * 5 = 25 \equiv 3$$

2. Calculate $1/8 \bmod 11$ (the inverse of 8) by hand using the equation subtracting algorithm. Use your result to calculate $5/8 \bmod 11$.

$$11x = 11$$

$$8x = 1$$

$$3x = 10$$

$$8x = 1$$

$$3x = 10$$

$$5x = -9 \equiv 2$$

$$5x = 2$$

$$3x = 10$$

$$2x = -8 \equiv 3$$

$$3x = 10$$

$$2x = 3$$

$$x = 7$$

$$\text{check: } 8 * 7 = 56 \equiv 1$$

$$5/8 = 5 * 7 = 35 \equiv 2.$$

Lecture 13

1. *How is polynomial arithmetic similar and how does it differ from using integers mod n ? In what way is it better? When must integers mod n be used rather than polynomial arithmetic?*

Polynomial arithmetic has the concept of a prime polynomial. All polynomials except 0 will have an inverse if their arithmetic is mod a prime (irreducible) polynomial. Polynomial arithmetic where the coefficients are integers mod 2 is more convenient than integers mod n because bitstrings can be converted directly into polynomials. Prime numbers are not powers of 2, and so bit strings need to be converted to integers before integers mod n , where n is not a power of 2, are used.

2. *Convert 1010_2 and 110_2 to polynomials, add them up, and convert the answer back to bit strings.*

$x^3 + x$, and $x^2 + x$. Adding them gives $x^3 + x^2 + 2x$, but $2x = 0$, so the addition gives $x^3 + x^2$. This corresponds to 1100_2 .

3. *Now multiply them, using irreducible polynomial $x^4 + x + 1$*

$$P = (x^3 + x) * (x^2 + x) = x^5 + x^4 + x^3 + x^2.$$

Remove x^5 term $Q = P - x * IP$.

$$Q = x^5 + x^4 + x^3 + x^2 - (x^5 + x^2 + x) = x^4 + x^3 + x.$$

Remove x^4 term $R = Q - IP$.

$$R = x^4 + x^3 + x - (x^4 + x + 1) = x^3 + 1 = 1001_2.$$

Lecture 15

1. Define the term "The entropy of a set of messages" and show how it can be calculated. A language contains 5 symbols: A, B, C, D and E. A, B, C each occur $\frac{1}{4}$ of the time, while D and E occur $\frac{1}{8}$ of the time. What is the entropy of this language?

$$\begin{aligned} & 3 * \frac{1}{4} * 2 + 2 * \frac{1}{8} * 3 \\ & = \frac{3}{2} + \frac{3}{4} \\ & = \frac{9}{4} \end{aligned}$$

2. Define the term "unicity distance." What information is needed to calculate it, and how useful is the concept of unicity distance? A newly invented language has 16 different symbols in its alphabet and is quite precise. On average each letter in the alphabet conveys 2 bits of information. A message in this language is encrypted with an 8 character key. It is known that users will choose English language keys all in lower case. What is the unicity distance of these encrypted messages?

$$\begin{aligned} \text{unicity distance} &= H(K) / D \\ H(K) &= 8 * 1.5 \text{ (r for English)} = 12 \\ D &= R - r = 4 - r = 2 \end{aligned}$$

So unicity distance = $12 / 2 = 6$.